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Master
Project Management
and Data Science

htw

Setting Up Python

Master's student Tales Thury



What You Will Have After This Guide

VS Code

Professional code editor, used by developers worldwide

Virtual Environment

An isolated workspace, your libraries stay organised

Miniconda

Manages Python and all your data science libraries

Jupyter in VS Code

Same cells you know from Colab, on your computer

All Steps at a Glance

1. Install VS Code

2. Install Python + Jupyter Extensions

3. Install Miniconda

4. Create Your Project Folder

5. Create a Virtual Environment

6. Open Folder in VS Code

7. Create a Jupyter Notebook

8. Select Your Environment as Kernel

9. Test That Everything Works

Install VS Code



Install VS Code

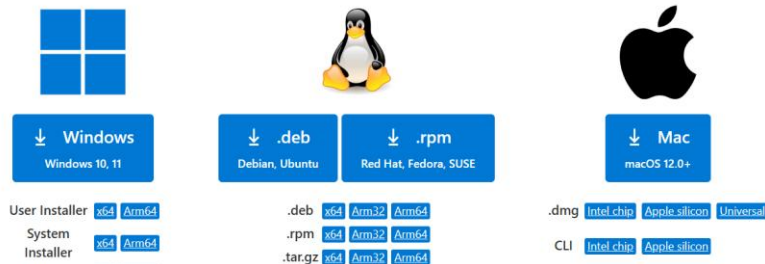
Go to:

<https://code.visualstudio.com/download>

- Choose your operating system: Windows, Mac, or Linux
- Download the installer and run it like any normal program
- Accept all default settings during installation

Download Visual Studio Code

Free and built on open source. Integrated Git, debugging and extensions.



The download page layout features three main columns. The first column is for Windows, with a Windows logo icon, a 'Windows' button (Windows 10, 11), and links for 'User Installer' and 'System Installer', each with 'x64' and 'Arm64' options. The second column is for Linux, with a Tux penguin icon, buttons for '.deb' (Debian, Ubuntu) and '.rpm' (Red Hat, Fedora, SUSE), and links for '.deb', '.rpm', and '.tar.gz', each with 'x64', 'Arm32', and 'Arm64' options. The third column is for Mac, with an Apple logo icon, a 'Mac' button (macOS 12.0+), and a link for '.dmg' with 'Intel chip', 'Apple silicon', and 'Universal' options, plus a 'CLI' link with 'Intel chip' and 'Apple silicon' options.

Operating System	Download Link / Package	Architecture Options
Windows	User Installer	x64, Arm64
	System Installer	x64, Arm64
Linux	.deb (Debian, Ubuntu)	x64, Arm32, Arm64
	.rpm (Red Hat, Fedora, SUSE)	x64, Arm32, Arm64
	.tar.gz	x64, Arm32, Arm64
	.deb	x64, Arm32, Arm64
Mac	.dmg	Intel chip, Apple silicon, Universal
	CLI	Intel chip, Apple silicon
	Mac	macOS 12.0+

Install Python + Jupyter Extensions



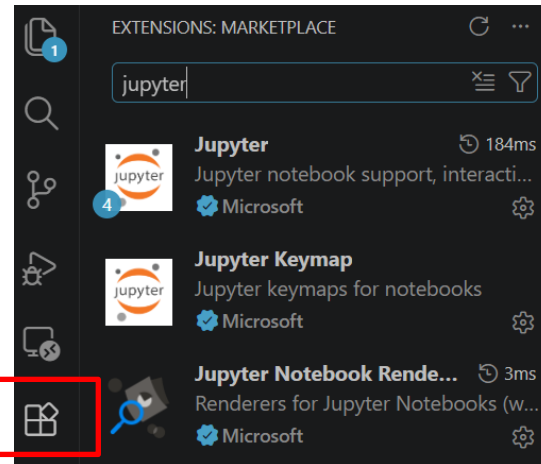
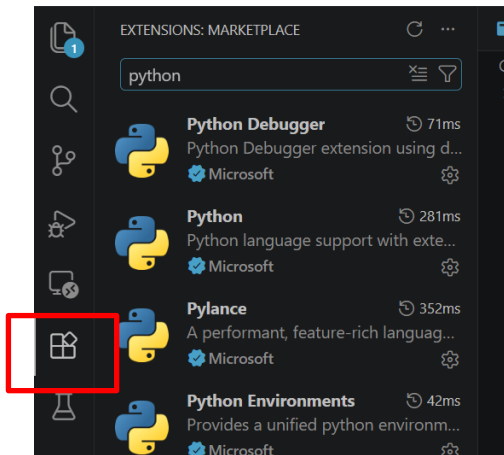
You need TWO extensions. Without both, notebooks will not work.

Python Extension - Teaches VS Code to understand Python files, syntax highlighting, error detection, autocomplete.

- Click the Extensions icon (four squares, left sidebar)
- Search for Python
- Install the one made by Microsoft

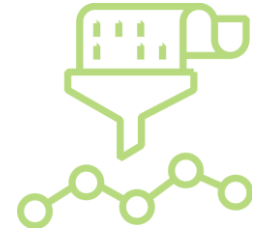
Jupyter Extension - Allows VS Code to open and run .ipynb notebook files, the same format as Colab.

- In the same Extensions panel
- Search for Jupyter
- Install the one made by Microsoft





Download and install Miniconda



Install Miniconda

Why Miniconda?

Miniconda installs Python and a tool called conda that manages your libraries and environments cleanly. Without it, installing libraries gets messy fast.

Miniconda is how you install and manage them safely.

Go to: <https://www.anaconda.com/docs/getting-started/miniconda/install/overview>

Windows

Windows guides



Graphical installer

Install Miniconda using a point-and-click interface. Recommended for users new to command-line tools.

Mac

macOS guides



Graphical installer

Install Miniconda using a standard macOS `.pkg` installer. Quick setup for users who prefer GUI workflows.

Tip: After installation, close and reopen any terminal. This loads the new conda commands.

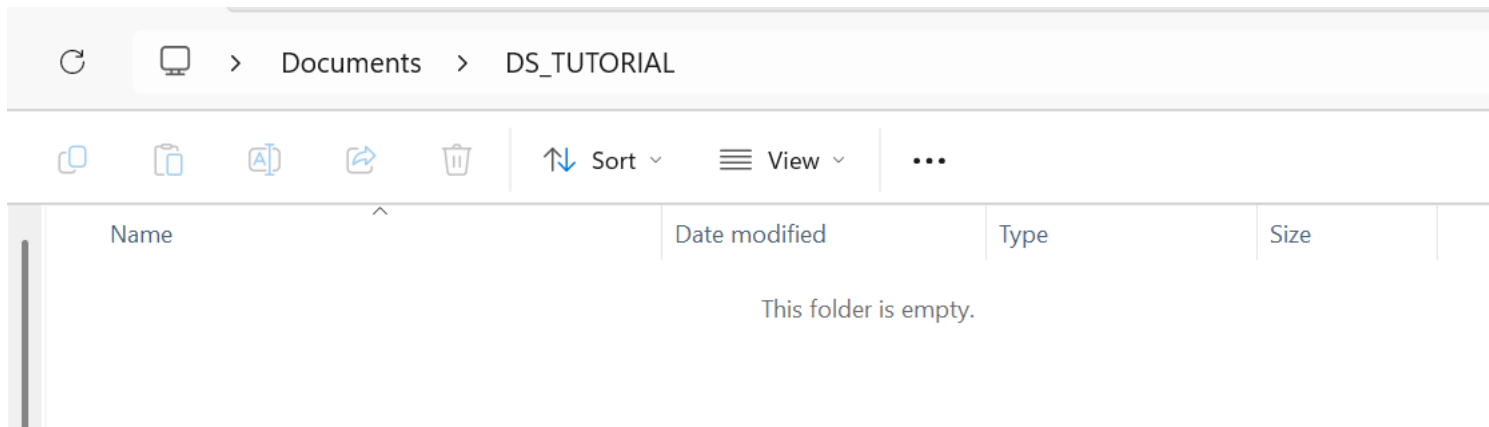
Create Folder + Virtual Environment





Folder and Virtual Environment

- Go to Documents and create a new folder called **DS_TUTORIAL**
- Avoid spaces and special characters in the folder name



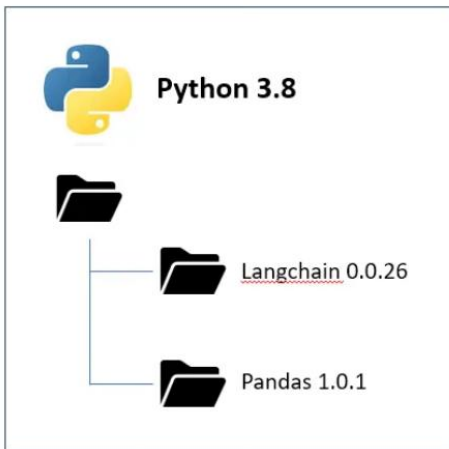


What is a virtual environment?

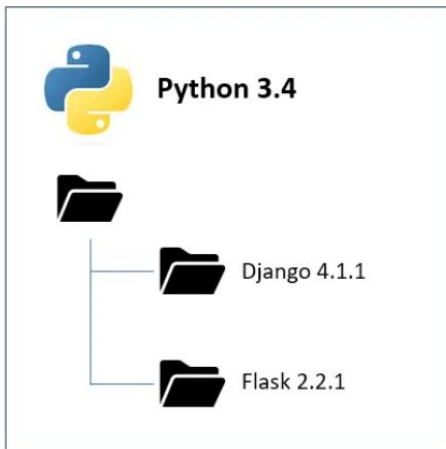
A virtual environment is a self-contained box on your computer that holds:

- a specific version of Python
- a specific set of libraries

Virtual Environment 1



Virtual Environment 2



Folder and Virtual Environment

Windows — Anaconda Prompt

```
cd C:\Users\YourName\Documents\DS_TUTORIAL  
  
conda create -n myenv python=3.11 pandas  
matplotlib seaborn scikit-learn jupyter
```

Mac — Terminal

```
cd /Users/YourName/Documents/DS_TUTORIAL  
  
conda create -n myenv python=3.11 pandas  
matplotlib seaborn scikit-learn jupyter
```

When it asks `Proceed ([y]/n)?` type `y` and press Enter

Important: This downloads several packages. It can take 3-5 minutes. Do not close the window.

How to Know Which Environment You Are In

Windows — Anaconda Prompt

```
(base) C:\Users\Tales> WRONG -  
base environment  
(myenv) C:\Users\Tales> CORRECT -  
myenv is active
```

Mac — Terminal

```
(base) tales@MacBook ~ WRONG  
(myenv) tales@MacBook ~ CORRECT
```

If you see (base) instead of (myenv) — run this first:

```
conda activate myenv
```

Important: If you run code in the wrong environment, pandas will not be found, even though you installed it. Always check (myenv) is showing before you start.

Task	Where to run it
<code>conda activate myenv</code>	Anaconda Prompt / Mac Terminal
<code>conda install pandas</code>	Anaconda Prompt / Mac Terminal
<code>pip install polars</code>	VS Code Terminal (with myenv active)



Anaconda Prompt

Check Conda Environments

Conda env list

```
(base) C:\Users\TalesThury>conda env list
# conda environments:
#
base                * C:\ProgramData\Anaconda3
myenv               C:\Users\TalesThury\.conda\envs\myenv
```

conda activate myenv

```
(base) C:\Users\TalesThury>conda activate myenv

(myenv) C:\Users\TalesThury>|
```

VS Code Open, Create, Connect



VS Code Setup

Open Folder

- File > Open Folder
- Select your **DS_TUTORIAL** folder
- Click 'Yes, I trust the authors'

Create Notebook

- Press Ctrl+Shift+P (Win) or Cmd+Shift+P (Mac)
- Type: create notebook
- Select: New Jupyter Notebook
- Save as tutorial.ipynb

Select Kernel

- Top-right corner
- Click 'Select Kernel'
- Choose:
- myenv (Python 3.11)



What Is a Kernel?

- The notebook is the car. The kernel is the engine under the hood.
- If you pick the wrong kernel, pandas does not exist — because that Python has no pandas installed
- Always select myenv as the kernel when working on this course

Wrong kernel selected:

```
ModuleNotFoundError: No module named  
'pandas'
```

Correct kernel (myenv):

```
import pandas as pd    # works perfectly
```

Important: If you get ModuleNotFoundError: click Select Kernel (top right) and choose myenv.

Run This in Your First Cell

Copy this into the first cell of your notebook and press Shift + Enter:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

print('Everything works!')
print('pandas version:', pd.__version__)
```

If you see Everything works! you are done.

Tip: Save this cell. Run it every time you open VS Code to confirm the environment is working.

If you see an error, check that myenv is selected as the kernel (Step 8).

Installing New Libraries in the Future

What if you want to install polars, tensorflow, or any other library later?

The library goes into whichever environment is active when you install it.

```
Step 1 – activate your environment first  
conda activate myenv
```

```
# Step 2 – install with conda (try this first)  
conda install polars
```

```
# If conda cannot find it, use pip  
pip install polars
```

- Try conda install first — it checks compatibility with other libraries
- Use pip install if conda cannot find the library
- Always make sure myenv is active before installing anything

```
conda list          # see everything installed in the current environment
```

Starting a New Project - Step by Step

1

Create a project folder

File Explorer / Finder

Create a new folder with a clear name, no spaces

`Documents/DS_TUTORIAL/MyProject`

2

Navigate to the folder

VS Code

File > Open Folder > select your folder

3

Check the kernel

VS Code

Top right corner, must show myenv (Python 3.11)

4

Create or open your notebook

VS Code

New file → name it something meaningful → start coding

When does a project end?

Start a new folder when

- new assignment,
- new dataset,
- new audience,
- conflicting libraries

Stay in the same folder when

- more analysis on same data,
- splitting cleaning + modelling,
- adding a new plot,
- trying two approaches to the same problem